

9.7 Volume in the Real World

Lesson Introduction

 Benchmark	MA.912.GR.4.5 Solve mathematical and real-world problems involving the volume of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres.			 Learning Target	I can use volume to solve problems with real-world applications.
 Benchmark Coherence	Building On			Working Toward	
	MA.6.GR.2.3 Solve mathematical and real-world problems involving the volumes of right rectangular prisms with positive rational-number edge lengths using a visual model and a formula. MA.7.GR.2.3 Solve mathematical and real-world problems involving volumes of right circular cylinders.			N/A	
 Essential Questions	- How is volume used in the real-world? - How can I use the appropriate formula to find the volume of a three-dimensional figure? - How can I use volume to solve problems with real-world applications?				
 Lesson Narrative	Lesson 7 adds to Lesson 6 by specifically having students work with real-world scenarios. By the start of this lesson, students should know the basic volume formulas and know how to substitute into a volume formula. Students work through a variety of real-world concepts to find volume or other needed information.				
 Component Summary	9.7.1 Warm-Up (~5 min.)	Finding and explaining the error in an advertisement (<i>SE p. 89</i>)	9.7.3 Guided Practice (5-10 min.)	Solving problems involving volume with guidance (<i>SE p. 90</i>)	
	9.7.2 Collaboration (10-15 min.)	Determining which packaging is best for a basketball (<i>SE p. 89</i>)	9.7.4 Your Turn! (10-15 min.)	Solving problems involving volume independently (<i>SE p. 91</i>)	
	9.7.5 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 93</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		(Optional) Basketball, Cardboard Boxes, Coffee Mug, Toy Truck, Tennis Ball Canister		N/A

 Teacher Tips	Common Misconceptions • Students may incorrectly substitute values into formulas or use incorrect formulas. • On 9.7.4 Your Turn!, students may incorrectly find the volume of one tennis ball and/or the volume of the air. • Students may forget to find the area of the base when finding the volume of a pyramid.	Things to Consider • On question 1 in 9.7.4 Your Turn!, consider having a canister of tennis balls to model the question. • Consider allowing students to work in partners on 9.7.4 Your Turn! if time runs short.
	Literacy and Language Routines Take Turns Have students take turns working various steps of the questions. Students justify their work and settle disagreements if those arise. Encourage students to paraphrase what their partner says throughout the lesson.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.7.1: Real-World Contexts This lesson provides students with multiple opportunities to engage with real-world concepts by connecting math with real-world experiences and to use models to solve questions. The questions provided in this lesson give students a variety of real-world scenarios and require various levels of thinking.
	 Differentiate Instruction Consider having models or manipulatives for this lesson. Find real-world objects that model a rectangular prism, cone, pyramid, sphere, and cylinder to have available for students to look at and use in class if necessary. Some examples are listed below. • Rectangular prism: any box • Cone: ice cream cones • Pyramid: paperweight, print a net and assemble • Sphere: basketball, baseball • Cylinder: paper towel roll Providing tangible objects for students to reference can help students better visualize volume and can also help students see volume in the real-world.	
Lesson Notes:		

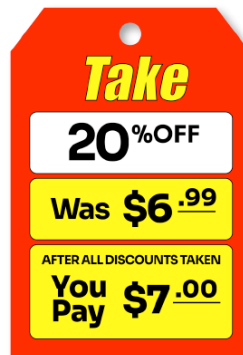
9.7.1 Warm-Up: Math Fail

Component Overview: Students examine a picture and determine what math mistake is in the picture and how to fix the issue.



9.7.1 Warm-Up: Math Fail

1. The clearance sign shown is what is known as a “math fail.”



Consider having students quickly share their responses with classmates. Allow students to share anywhere they may have seen math errors like this in the real-world before. Discuss the financial impact this error could have on a company.

- a. What do you notice about the sign that makes this a math fail?

Answers will vary.

What is being asked to pay is more than the original price.

- b. Write a letter to the company explaining why this is a math fail and how to fix the issue.

Answers will vary.

This is a math fail because the total cost after the discount should not be more than the original price. Whoever wrote the discount tag should have taken 20% off of the original price of \$6.99 to get the new price to pay of \$5.60.

9.7.2 Collaboration: Packing Peanuts

Component Overview: 9.7.2 Collaboration provides students with an opportunity to engage with a real-world scenario to figure out which type of packaging is best for a basketball with a given diameter. Students find the volume of a rectangular prism, cylinder, and cube to answer the question by determining which packaging is best.

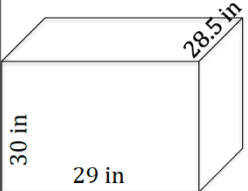
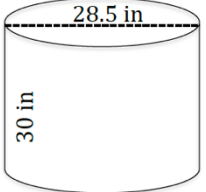
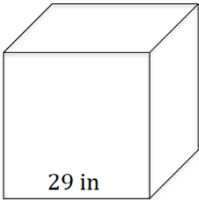


9.7.2 Collaboration: Packing Peanuts

1. Josiah works for a sporting goods company in the shipping department. Josiah is trying to determine the best solution for packing a basketball with a diameter of 28.5 inches.
 - a. What is the volume of the basketball?

$$V = \frac{4}{3}\pi(14.25)^3$$

$$V = 12,120.854 \text{ inches}^3$$
 - b. What size box do you think would be the best fit?
Answer will vary.
 - c. Find the volume of the three options of boxes below, from which Josiah can choose.

Rectangular Prism		Volume = <u>24,795 inches³</u>
Cylinder		Volume = <u>19,138.190 inches³</u>
Cube		Volume = <u>24,389 inches³</u>



Have students make a prediction as to which type of packaging would be the best fit and share with classmates to hear a variety of predictions.



Consider using a jigsaw strategy by assigning students either the rectangular prism, cylinder, or cube, and have students find the volume of their assigned figure. Bring the class together and have all groups share their volumes and have students record the results.

9.7.2 Collaboration: Packing Peanuts (continued)

- d. Discuss with your partner how to determine the amount of space left over in the package after the basketball has been placed in it. Summarize your discussion.

Answers will vary.

Get the volume of the outer box being used and the volume of the basketball and subtract from each other to find the amount of space left over inside the package.



Have students share discussions with other groups or with the class before beginning question 1e.

- e. After the basketball has been placed in the package, the remaining space is filled with packing peanuts. If each peanut is 1 cm^3 , which box will use the least amount of packing peanuts?

The difference between the volume of the cylindrical box and the volume of the basketball is the smallest, so the cylindrical box will need the least number of peanuts.

9.7.3 Guided Practice: Volume

Component Overview: Students complete two practice questions on finding the volume of a rectangular prism and two cylinders.



9.7.3 Guided Practice: Volume

- Keon is building a raised garden bed that will have a width of 3.5 feet (ft), a depth of 2 ft, and length of 3 ft. If Keon fills the garden bed with soil, how much soil does he need to purchase?

$$V = l \cdot w \cdot h$$

$$V = (3.5)(2)(3)$$

$$V = 21 \text{ ft}^3$$

Keon needs to purchase 21 ft^3 of soil.

- Jasper wants to buy a new coffee mug, and he has narrowed it down to two options. The dimensions are given in inches (in.)

Option A	Option B
 5.25 in Diameter: 2 in	 5 in Radius: 1.25 in

- What is the volume, rounded to the thousandths place, of option A?

$$V = \pi(1^2)(5.25)$$

$$V = 16.493 \text{ in}^3$$

- What is the volume, rounded to the thousandths place, of option B?

$$V = \pi(1.25^2)(5)$$

$$V = 24.544 \text{ in}^3$$



On question 2, the handles are not part of the diagram. Remind students to not include the handle when finding the volume.



In the event of a time crunch, assign half of the class Option A and assign the other half Option B.

9.7.4: Your Turn!

Component Overview: Students work through a variety of application problems relating to finding the volume of a three-dimensional figure. On question 1, students find the volume of a canister and the air that is left over. In question 2, students compare the volumes of a sphere and a cone and must answer a statement. Students find the volume of a truck in question 3 and question 4 asks students to find the volume of a pyramid.



9.7.4 Your Turn!

1. A tennis ball canister with three tennis balls is shown. The diameter of the canister is 2.7 inches (in) and the height is 8.1 in.

a. What is the volume of the canister?

$$V = \pi(1.35^2)(8.1)$$

$$V = 46.377 \text{ in}^3$$

b. What is the volume of 1 tennis ball?

$$V = \frac{4}{3}\pi(1.35)^3$$

$$= 10.306 \text{ in}^3$$

c. How much air is left in the canister after the tennis balls have been placed inside?

$$V = \text{total volume} - (\text{volume of 3 tennis balls})$$

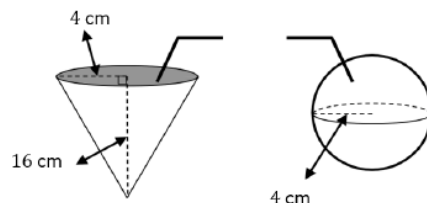
$$= 46.377 - 3(10.306)$$

$$= 15.459 \text{ in}^3$$



Consider allowing students to continue working with partners and/or groups on this component of the lesson.

2. Club Cool in Epcot offers free samples of Coke products that are distributed worldwide. Flavors include Fanta Pineapple, Fanta Melon, Inca Cola, VegitaBeta, and Sparberry. You can choose to have your sample served in either of the containers shown.



Common Misconception

Students may think the answer to question 1c is the answer to question 1b.

9.7.4: Your Turn! (continued)

Gian, Yanessa, and Jack made different conclusions about the drink containers. Who is correct and why?

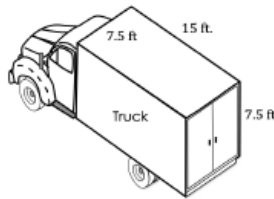
Gian: The volumes are the same because the height of the cone is 4 times its radius, and both containers have the same radius.

Yanessa: The volume of the sphere is greater than the cone's because $\frac{4}{3}\pi r^3$ is greater than $\frac{1}{3}\pi r^2 h$.

Jack: The volume of the cone is greater than the sphere's because the height is 4 times the radius of both containers.

Gian is correct because the height of the cone is 4 times its radius, and both containers have the same radius.

3. Julian owns a moving company and has a truck that measures 15 ft $\times$ 7.5 ft $\times$ 7.5 ft.



He charges individuals by the number of boxes in the moving truck. If each box is a cube with side lengths of 1.5 ft, how many boxes can he fit in the truck?

$$V = l \cdot w \cdot h$$

$$V = (7.5)(15)(7.5)$$

$$V = 843.75 \text{ ft}^3$$

$$\text{Volume of the truck} = 843.75 \text{ ft}^3$$

$$V = a^3$$

$$V = 1.5^3$$

$$V = 3.375 \text{ ft}^3$$

$$\text{Volume of the box} = 3.375 \text{ ft}^3$$

$$\frac{843.75}{3.375} = 250 \text{ boxes}$$



Tell students that the figures coming out of the cone and sphere are representing drinking straws and are not needed to solve the question.

9.7.4: Your Turn! (continued)

4. The Pyramid of Cestius, pictured below, is located in Rome, Italy. The base of the pyramid is a square with side lengths of 30 meters and stands at a height of 36.5 meters. What is the volume of the pyramid?



$$V = \frac{1}{3}Bh$$

$$V = \frac{1}{3}(30)^2(36.5)$$

$$V = 10,950\text{m}^3$$



Common Misconception

Students may want to divide 843.75 by 1.5 to find the number of cubes. Monitor students' work to see if this occurs. If so, have students explain their thinking.



Consider asking the following for question 3.

- How can you find the volume of a box?
- How can you find the total number of boxes?



Remind students that when finding the volume of a pyramid, the B is the area of the base. Students may forget to square the side length of the base (30 meters) and not find the area of the square base.

9.7.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



9.7.5 Wrap-Up: Cool Down

1. Provide a real-world context that would require you to find the volume of a three-dimensional figure.

Answers will vary.